

```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

data=pd.read_csv('salary_data.csv')

data.head()

```

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0

```

data.shape

(30, 2)

data.isnull().sum()

YearsExperience    0
Salary            0
dtype: int64

x=data.iloc[:,1].values
y=data.iloc[:,2].values

x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.2,
random_state=42)

model=LinearRegression()
model.fit(x_train, y_train)
y_pred=model.predict(x_test)

train_accuracy = model.score(x_train, y_train)
test_accuracy = model.score(x_test, y_test)

print("Training Accuracy (R² Score):", train_accuracy)
print("Testing Accuracy (R² Score):", test_accuracy)

Training Accuracy (R² Score): 0.9645401573418146
Testing Accuracy (R² Score): 0.9024461774180497

print(y_pred)
print(y_test)

[[115790.21011287]
 [ 71498.27809463]
[102596.86866063]
 [ 75267.80422384]

```

```
[ 55477.79204548]
[ 60189.69970699]]
[[112635.]
 [ 67938.]
 [113812.]
 [ 83088.]
 [ 64445.]
 [ 57189.]]
```

```
plt.scatter(x_train, y_train, color='blue')
plt.plot(x_train, model.predict(x_train), color='red')
plt.title('SALARY VS EXPERIENCE (training set)')
plt.xlabel('Experience in Years')
plt.ylabel('Salary in Rupees')
plt.show()
```



```
plt.scatter(x_test, y_test, color='blue')
plt.plot(x_train, model.predict(x_train), color='red')
plt.title('SALARY VS EXPERIENCE (testing set)')
plt.xlabel('Experience in Years')
plt.ylabel('Salary in Rupees')
plt.show()
```



```

import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error,
mean_squared_error

df = pd.read_csv("Taxi_Set.csv")

print(df.head())
print(df.shape)
print(df.columns)

```

	trip_duration_sec	trip_duration_min	trip_duration_hr \
0	748	12.47	0.207833
1	1187	19.78	0.329667
2	730	12.17	0.202833
3	671	11.18	0.186333
4	329	5.48	0.091300

	distance_traveled_Km	KPH	wait_time_cost	distance_cost \
0	2.75	13.231756	4.97	4.27
1	3.43	10.404449	7.89	5.33
2	3.12	15.382087	4.85	4.85
3	5.63	30.214669	4.46	8.75
4	2.09	22.883212	2.19	3.25

	fare_w_flag	tip	miscellaneous_fees	total_fare_new
0	11.94	24	6.30	42.24
1	15.92	24	13.20	53.12
1	12.40	0	26.63	39.03
2	15.91	0	9.75	25.66
3	8.14	12	13.20	33.34
4				

	surge_applied
0	False
1	False
2	True
3	False
4	False

```

(207596, 13)
Index(['trip_duration_sec', 'trip_duration_min', 'trip_duration_hr',
      'distance_traveled_Km', 'KPH', 'wait_time_cost',

```

```
'distance_cost',
    'fare_w_flag', 'tip', 'miscellaneous_fees', 'total_fare_new',
    'num_of_passengers', 'surge_applied'],
    dtype='object')
```

```
df = df.dropna()
```

```
print(df.isnull().sum())
```

```
trip_duration_sec      0
trip_duration_min      0
trip_duration_hr       0
distance_traveled_Km   0
KPH                    0
wait_time_cost         0
distance_cost          0
fare_w_flag            0
tip                    0
miscellaneous_fees     0
total_fare_new         0
num_of_passengers      0
surge_applied          0
dtype: int64
```

```
X = df.drop(columns=["total_fare_new"])
```

```
y = df["total_fare_new"]
```

```
print(X.head())
```

	trip_duration_sec	trip_duration_min	trip_duration_hr \
0	748	12.47	0.207833
1	1187	19.78	0.329667
2	730	12.17	0.202833
3	671	11.18	0.186333
4	329	5.48	0.091300

	distance_traveled_Km	KPH	wait_time_cost	distance_cost \
0	2.75	13.231756	4.97	4.27
1	3.43	10.404449	7.89	5.33
2	3.12	15.382087	4.85	4.85
3	5.63	30.214669	4.46	8.75
4	2.09	22.883212	2.19	3.25

	fare_w_flag	tip	miscellaneous_fees	num_of_passengers
surge_applied				
0	11.94	24	6.30	1
False				
1	15.92	24	13.20	1
False				
2	12.40	0	26.63	1
True				

3	15.91	0	9.75	3
False				
4	8.14	12	13.20	1
False				

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
print("Training Data:", X_train.shape)
print("Testing Data :", X_test.shape)
```

```
Training Data: (166076, 12)
Testing Data : (41520, 12)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
print("Model trained successfully!")
```

```
Model trained successfully!
```

```
train_pred = model.predict(X_train)
test_pred = model.predict(X_test)
```

```
train_accuracy = r2_score(y_train, train_pred)
test_accuracy = r2_score(y_test, test_pred)
```

```
print("Training Accuracy :", round(train_accuracy,4))
print("Testing Accuracy :", round(test_accuracy,4))
```

```
Training Accuracy : 1.0
Testing Accuracy : 1.0
```

```
mae = mean_absolute_error(y_test, test_pred)
rmse = np.sqrt(mean_squared_error(y_test, test_pred))
```

```
print("MAE :", round(mae,2))
print("RMSE:", round(rmse,2))
```

```
MAE : 0.0
RMSE: 0.0
```

```
sample = pd.DataFrame([
    "trip_duration_sec":900,
    "trip_duration_min":15,
    "trip_duration_hr":0.25,
    "distance_traveled_Km":10,
```

```
    "KPH":40,  
    "wait_time_cost":20,  
    "distance_cost":150,  
    "fare_w_flag":180,  
    "tip":20,  
    "miscellaneous_fees":15,  
    "num_of_passengers":2,  
    "surge_applied":1  
})  
  
prediction = model.predict(sample)  
  
print("Predicted Fare:", prediction[0])  
  
Predicted Fare: 212.566666666667846
```

```
import pandas as pd
data = {
    'Weight': [79, 69, 73, 95, 82, 55, 69, 71, 64, 69],
    'Height': [1.80, 1.68, 1.82, 1.70, 1.87, 1.55, 1.50, 1.78, 1.67,
1.64],
    'Age': [35, 39, 25, 60, 27, 18, 89, 42, 16, 52],
    'Gender': ['Male', 'Male', 'Male', 'Male', 'Male',
    'Female', 'Female', 'Female', 'Female', 'Female']
}
df = pd.DataFrame(data)
df.to_csv('body_weight.csv', index=False)
print("body_weight.csv created successfully!")
print(df)
```

body_weight.csv created successfully!

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male
5	55	1.55	18	Female
6	69	1.50	89	Female
7	71	1.78	42	Female
8	64	1.67	16	Female
9	69	1.64	52	Female

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

```
data = pd.read_csv('body_weight.csv')
```

```
data.head()
```

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male

```
data.shape
```

```
data.isnull().sum()
```

```
Weight      0
Height      0
Age         0
Gender      0
dtype: int64
```

```
data['Gender'] = data['Gender'].map({'Male':1, 'Female':0})
```



```

x = data[['Height', 'Age', 'Gender']]
y = data[['Weight']]

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(x_train, y_train)

LinearRegression()

y_pred = model.predict(x_test)
print("Predicted Weight:")
print(y_pred)
print("\nActual Weight:")
print(y_test.values)

Predicted Weight:
[[56.85651025]
 [79.10576115]]

Actual Weight:
[[64]
 [69]]

train_accuracy = model.score(x_train, y_train)
test_accuracy = model.score(x_test, y_test)
print("Training Accuracy:", train_accuracy)
print("Testing Accuracy:", test_accuracy)

Training Accuracy: 0.8312942630906153
Testing Accuracy: -11.252468343750815

```